APEX GLOBAL BANK

SR 11-7 MODEL DEVELOPMENT DOCUMENT

FRTB Standardised Approach

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Owner	Dr. Priya Nair (CRO)
Validator	Dr. Samuel Achebe (Model Validation Officer)
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Regulatory Framework: BCBS d352 (Jan 2016), CRR3 2024, SR 11-7

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Contents

1 Model Overview **3**
1.1 Purpose . . . 3
1.2 Business Use . . . 3

2 Theoretical Basis **3**
2.1 Three Capital Components . . . 3
2.1.1 1. Sensitivity-Based Method (SBM) . . . 3
2.1.2 2. Residual Risk Add-On (RRAO) . . . 3
2.1.3 3. Default Risk Charge (DRC) . . . 3
2.2 IR Tenor Buckets (GIRR) . . . 4

3 Mathematical Specification **4**
3.1 Equity Risk Weights by Bucket . . . 4
3.2 IR Risk Weights (GIRR) . . . 4

4 Implementation **4**

5 Validation **4**

6 Model Limitations **4**

7 Open Findings **5**

1 Model Overview

1.1 Purpose

The FRTB Standardised Approach (SA) replaces the Basel 2.5 Standardised Approach with a sensitivity-based method (SBM) that better reflects risk. Under Basel IV / CRR3, all banks must report FRTB-SA capital; IMA banks use it as a floor. Apex uses FRTB-SA as the primary capital model pending IMA approval.

1.2 Business Use

- **Regulatory capital (primary):** SA capital is the binding constraint until IMA approval.
- **Internal limit calibration:** SA sensitivities inform risk appetite for delta, vega, curvature.
- **Parallel run:** Compared against IMA output; material divergence triggers governance review.

2 Theoretical Basis

2.1 Three Capital Components

2.1.1 1. Sensitivity-Based Method (SBM)

Capital is computed from risk factor sensitivities across 7 risk classes (GIRR, CSR non-sec, CSR sec, FX, EQ, Commodity, VEGA).

Delta capital:

$$K_\delta = \sqrt{\sum_b \sum_{i,j \in b} \rho_{ij}\, WS_i\, WS_j}$$

where $WS_i = RW_i \times s_i$ (risk weight $\times$ net sensitivity), ρ_{ij} is the intra-bucket correlation.

Cross-bucket aggregation:

$$K_{\text{SBM}} = \sqrt{\sum_b K_b^2 + \sum_{b \neq c} \gamma_{bc}\, S_b\, S_c}$$

where γ_{bc} is the cross-bucket correlation and $S_b = \sum_i WS_i$ within bucket b.

2.1.2 2. Residual Risk Add-On (RRAO)

$$\text{RRAO} = 1.0\% \times N_{\text{exotic}} + 0.1\% \times N_{\text{other}}$$

Applies to instruments with path-dependent payoffs, correlation products, and non-linear exotic optionality.

2.1.3 3. Default Risk Charge (DRC)

$$\text{DRC} = \max\Big(0,\ \sum_i \text{JTD}_i \times \text{LGD}_i \times RW_i\Big) \times W_{\text{maturity}}$$

Jump-to-default exposure for non-securitisation credit positions.

2.2 IR Tenor Buckets (GIRR)

0.25Y, 0.5Y, 1Y, 2Y, 3Y, 5Y, 10Y, 15Y, 20Y, 30Y (10 buckets per currency).

3 Mathematical Specification

3.1 Equity Risk Weights by Bucket

Bucket	Description	Delta RW
1	Large-cap EM	55%
8	Large-cap DM	25%
11	Small-cap	70%

3.2 IR Risk Weights (GIRR)

Tenor-dependent: 1.7% (3m) to 1.5% (10Y), prescribed by BCBS d352 §21.52.

Correlation parameters: Three scenarios (high/medium/low) for conservative capital: $\rho =$ 1.0×, 0.75×, 0.5× of base correlations.

4 Implementation

Code location: `infrastructure/risk/regulatory_capital.py` — SA RWA calculator.

Sensitivities source: `GreeksCalculator` provides delta and vega. Curvature:

$$CVR_i = -V(S_i + RW_i S_i) + V(S_i - RW_i S_i) - 2V(S_i)$$

Capital aggregation: Three correlation scenarios; capital = max across scenarios.

5 Validation

Benchmark: FRTB-SA capital benchmarked against ISDA SIMM for non-cleared derivatives. Material divergence (>20%) triggers a model review.

Sensitivity validation: Delta sensitivities validated against analytic Greeks for vanilla equity options.

6 Model Limitations

1. **Curvature proxy:** CVR computed analytically using bump-and-reprice; no full Monte Carlo revaluation.
2. **RRAO classification:** Exotic optionality identification is manual; no automated product classifier.

3. **DRC recovery assumptions:** Recovery rate 40% applied uniformly; actual recovery varies by seniority.

7 Open Findings

ID	Severity	Description	Status
FRTB-F1	**Major**	Curvature risk aggregation not validated against independent vendor benchmark.	Open
FRTB-F2	**Major**	RRAO treatment for path-dependent exotics not documented; manual classification in use.	Open
FRTB-F3	**Minor**	DRC jump-to-default recovery rate uniformly 40%; seniority-specific recovery not modelled.	Open